

JEE MAINS & IIT JEE ADVANCED

MATHEMATICS

ELLIPSE

DIBY (DO IT BY YOURSELF)

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DIBY-01

The eccentricity of the ellipse $9x^2 + 5y^2 - 30y = 0$ is

- (A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{3}{4}$ (D) $\frac{5}{6}$

[Ans. : (B)]

DIBY-02

The co-ordinates of a focus of the ellipse $4x^2 + 9y^2 = 1$, is

- (A) $\left(\frac{\sqrt{5}}{6}, 0\right)$ (B) $\left(\frac{\sqrt{5}}{3}, 0\right)$ (C) $\left(\frac{\sqrt{5}}{4}, 0\right)$ (D) $\left(\frac{\sqrt{5}}{5}, 0\right)$

[Ans. : (A)]

DIBY-03

The latus rectum of the conic $3x^2 + 4y^2 - 6x + 8y - 5 = 0$ is

- (A) 3 (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{2}{\sqrt{3}}$ (D) $\frac{4}{\sqrt{3}}$

[Ans. : (A)]

DIBY-04

The sum of the distances of any point on the ellipse $3x^2 + 4y^2 = 24$ from its foci is

- (A) $8\sqrt{2}$ (B) $4\sqrt{2}$ (C) $16\sqrt{2}$ (D) 6

[Ans.: (B)]

DIBY-05

In an ellipse the distance between the foci is 8 and the distance between the directrices is 25, then the ratio of the length of major and minor axis is

- (A) $\frac{5}{\sqrt{17}}$ (B) $\frac{3}{\sqrt{17}}$ (C) $\frac{4}{\sqrt{17}}$ (D) $\frac{6}{\sqrt{17}}$

[Ans.: (A)]

DIBY-06

If the length of the minor axis of an ellipse is equal to one fourth of the distance between the foci, then the eccentricity of the ellipse is **[JEE Mains-2026]**

- (A) $\frac{4}{\sqrt{17}}$ (B) $\frac{\sqrt{3}}{16}$ (C) $\frac{3}{\sqrt{19}}$ (D) $\frac{\sqrt{5}}{7}$

[Ans.: (A)]

DIBY-07

If the co-ordinates of two points A and B are $(\sqrt{7}, 0)$ & $(-\sqrt{7}, 0)$ respectively and P is any point on the conic, $9x^2 + 16y^2 = 144$, then $PA + PB$ is equal to **[JEE Mains-2024]**

- (A) 9 (B) 16 (C) 6 (D) 8

[Ans. : (D)]

DIBY-08

An ellipse, with foci at $(0, 2)$ and $(0, -2)$ and minor axis of length 4, passes through which of the following points? **[JEE Mains-2019]**

- (A) $(1, 2\sqrt{2})$ (B) $(2, \sqrt{2})$ (C) $(2, 2\sqrt{2})$ (D) $(\sqrt{2}, 2)$

[Ans. : (D)]

DIBY-09

The length of the latus-rectum of the ellipse, whose foci are $(2, 5)$ and $(2, -3)$ and eccentricity is $\frac{4}{5}$, is **[JEE Mains-2024]**

- (A) $\frac{6}{5}$ (B) $\frac{50}{3}$ (C) $\frac{10}{3}$ (D) $\frac{18}{5}$

[Ans. : (D)]



DIBY-10

Let S and S' be the foci of an ellipse and B be any one of the extremities of its minor axis. If $\triangle S'BS$ is a right angled triangle with right angle at B and area $(\triangle S'BS) = 8$ sq units, then the length of a latus rectum of the ellipse is **[JEE Mains-2019]**

- (A) $2\sqrt{2}$ (B) $4\sqrt{2}$ (C) 2 (D) 4

[Ans. : (D)]

DIBY-11

Let C be the largest centred at $(2, 0)$ and inscribed in the ellipse $\frac{x^2}{36} + \frac{y^2}{16} = 1$. If $(1, \alpha)$ lies on C , then $10\alpha^2$ is equal to **[JEE Mains-2023]**

[Ans. : 118]

DIBY-12

The line $y = x + 1$ meets the ellipse $\frac{x^2}{4} + \frac{y^2}{2} = 1$ at two points P and Q . If r is the radius of the circle with PQ as diameter then $(3r)^2$ is equal to **[JEE Mains-2022]**

- (A) 20 (B) 12 (C) 11 (D) 8

[Ans. : (A)]



DIBY-13

Tangents are drawn to the ellipse $x^2 + 2y^2 = 2$ at all points on the ellipses then the locus of the mid-point of the intercept made by the tangents between the coordinates axes is

[IIT-JEE-2004]

- (A) $\frac{1}{2x^2} + \frac{1}{4y^2} = 1$ (B) $\frac{1}{4x^2} + \frac{1}{2y^2} = 1$ (C) $\frac{x^2}{2} + \frac{y^2}{4} = 1$ (D) $\frac{x^2}{4} + \frac{y^2}{2} = 1$

[Ans. : (A)]

DIBY-14

The number of values of c such that the straight line $y = 4x + c$ touches the curve

$\frac{x^2}{4} + y^2 = 1$ is

[IIT-JEE-1998]

- (A) 0 (B) 2 (C) 1 (D) ∞

[Ans. : (B)]

DIBY-15

Find the equation of the common tangents in 1st quadrant to the circle $x^2 + y^2 = 16$ and

$\frac{x^2}{25} + \frac{y^2}{4} = 1$. Also, find the length of the intercept of the tangent between the coordinates axes.

[IIT-JEE-2005]

[Ans. : $\frac{14}{\sqrt{3}}$]

DIBY-16

If the line $x - 2y = 12$ is tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2}$ at the point $\left(3, -\frac{9}{2}\right)$, then the length of the latus rectum of the ellipse is **[JEE Mains-2019]**

- (A) $12\sqrt{2}$ (B) 9 (C) $8\sqrt{3}$ (D) 5

[Ans.: (B)]

DIBY-17

Let a tangent to the curve $9x^2 + 16y^2 = 144$ intersect the coordinate axes at the points A and B . Then, the minimum length of the line segment AB is **[JEE Mains-2023]**

[Ans.:]

DIBY-18

If the tangent to the parabola $y^2 = x$ at a point (α, β) , ($\beta > 0$) is also a tangent to the ellipse $x^2 + 2y^2 = 1$, then α is equal to **[JEE Mains-2019]**

- (A) $2\sqrt{2} - 1$ (B) $2\sqrt{2} + 1$ (C) $\sqrt{2} + 1$ (D) $\sqrt{2} - 1$

[Ans.: (C)]

DIBY-19

Comprehension/Passage Based

Tangents are drawn from the point $P(3, 4)$ to the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ touching the ellipse at point A and B .
[IIT-JEE-2010]

(i) The coordinates of A and B are

(A) $(3, 0)$ and $(0, 2)$

(B) $\left(-\frac{8}{5}, \frac{2\sqrt{161}}{15}\right)$ and $\left(-\frac{9}{5}, \frac{8}{5}\right)$

(C) $\left(-\frac{8}{5}, \frac{2\sqrt{161}}{15}\right)$ and $(0, 2)$

(D) $(3, 0)$ and $\left(-\frac{9}{5}, \frac{8}{5}\right)$

[Ans. : (D)]

(ii) The orthocentre of the $\triangle PAB$ is

(A) $\left(5, \frac{8}{7}\right)$

(B) $\left(\frac{7}{5}, \frac{25}{8}\right)$

(C) $\left(\frac{11}{5}, \frac{8}{5}\right)$

(D) $\left(\frac{8}{25}, \frac{7}{5}\right)$

[Ans. : (C)]



DIBY-20

Let PQ be a focal chord of the parabola $y^2 = 4x$ such that it subtends an angle of $\frac{\pi}{2}$ at the point $(3, 0)$. Let the line segment PQ be also a focal chord of the ellipse $E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a^2 > b^2$. If e is the eccentricity of the ellipse E , then the value of $\frac{1}{e^2}$ is equal to **[JEE Mains-2022]**

- (A) $1 + \sqrt{2}$ (B) $3 + 2\sqrt{2}$ (C) $1 + 2\sqrt{3}$ (D) $4 + 5\sqrt{3}$

[Ans. : (B)]

DIBY-21

If two tangents drawn from a point (α, β) lying on the ellipse $25x^2 + 4y^2 = 1$ to the parabola $y^2 = 4x$ are such that the slope of one tangent is four times the other, then the value of $(10\alpha + 5)^2 + (16\beta^2 + 50)^2$ equals **[JEE Mains-2022]**

[Ans. : 2929]

DIBY-22

The acute angle between the pair of tangents drawn to the ellipse $2x^2 + 3y^2 = 5$ from the point $(1, 3)$ is **[JEE Mains-2022]**

- (A) $\tan^{-1} \left(\frac{16}{7\sqrt{5}} \right)$ (B) $\tan^{-1} \left(\frac{24}{7\sqrt{5}} \right)$ (C) $\tan^{-1} \left(\frac{32}{7\sqrt{5}} \right)$ (D) $\tan^{-1} \left(\frac{3+8\sqrt{5}}{35} \right)$

[Ans. : (B)]

DIBY-23

Let the common tangents to the curves $4(x^2 + y^2) = 9$ and $y^2 = 4x$ intersect at the point Q . Let an ellipse, centered at the origin O , has lengths of semi-minor and semi-major axes equal to OQ and 6, respectively. If e and l respectively denote the eccentricity and the length of the latus rectum of this ellipse, then $\frac{1}{e^2}$ is equal to

[JEE Mains-2022]

[Ans. : 4]

DIBY-24

Let θ be the acute angle between the tangents to the ellipse $\frac{x^2}{9} + \frac{y^2}{1} = 1$ and the circle $x^2 + y^2 = 3$ at their point of intersection in the first quadrant. Then $\tan\theta$ is equal to

[JEE Mains-2021]

(A) $\frac{5}{2\sqrt{3}}$

(B) 2

(C) $\frac{2}{\sqrt{3}}$

(D) $\frac{4}{\sqrt{3}}$

[Ans. : (C)]

DIBY-25

Consider the ellipse $\frac{x^2}{4} + \frac{y^2}{3} = 1$. Let $H(\alpha, 0)$, $0 < \alpha < 2$, be a point. A straight line drawn through H parallel to the y -axis crosses the ellipse and its auxiliary circle at points E and F respectively, in the first quadrant. The tangent to the ellipse at the point E intersects the positive x -axis at a point G . Suppose the straight line joining F and the origin makes an angle ϕ with the positive x axis.

[JEE (Adv.)-2022]

List-I		List-II	
(I)	If $\phi = \frac{\pi}{4}$, then the area of the triangle FGH is	(P)	$\frac{(\sqrt{3}-1)^4}{8}$
(II)	If $\phi = \frac{\pi}{3}$, then the area of the triangle FGH is	(Q)	1
(III)	If $\phi = \frac{\pi}{6}$, then the area of the triangle FGH is	(R)	$\frac{3}{4}$
(IV)	If $\phi = \frac{\pi}{12}$, then the area of the triangle FGH is	(S)	$\frac{1}{2\sqrt{3}}$
		(T)	$\frac{3\sqrt{3}}{2}$

[Ans. : (I) $\rightarrow$ (Q); (II) $\rightarrow$ (T); (III) $\rightarrow$ (S); (IV) $\rightarrow$ (P)]

DIBY-26

Let E be the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$. For any three distinct points P, Q and Q' on E , let $M(P, Q)$ be the mid-point of the line segment joining P and Q , and $M(P, Q')$ be the mid-point of the line segment joining P and Q' . Then the maximum possible value of the distance between $M(P, Q)$ and $M(P, Q')$ as P, Q and Q' vary on E , is _____.

[JEE (Adv.)-2021, Paper-1]

[Ans. : 4]

DIBY-27

Let the tangent and normal at the point $(3\sqrt{3}, 1)$ on the ellipse $\frac{x^2}{36} + \frac{y^2}{4} = 1$ meet the y axis at the points A and B respectively. Let the circle C be drawn taking AB as a diameter and the line $x = 2\sqrt{5}$ intersect C at the points P and Q . If the tangent at the points P and Q on the circle intersect at the point (α, β) , then $\alpha^2 - \beta^2$ is equal to

[JEE Mains-2023]

- (A) 61 (B) $304/5$ (C) 60 (D) $314/5$

[Ans. : (B)]

DIBY-28

Let the ellipse $E_1: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a > b$ and $E_2: \frac{x^2}{A^2} + \frac{y^2}{B^2} = 1, A < B$ have same eccentricity $\frac{1}{\sqrt{3}}$. Let the product of their lengths of latus rectums be $\frac{32}{\sqrt{3}}$, and the distance between the foci of E_1 be 4. If E_1 and E_2 meet at A, B, C and D , then the area of the quadrilateral $ABCD$ equals:

[JEE Mains-2025]

- (A) $6\sqrt{6}$ (B) $\frac{12\sqrt{6}}{5}$ (C) $\frac{24\sqrt{6}}{5}$ (D) $\frac{18\sqrt{6}}{5}$

[Ans. : (C)]

DIBY-29

The length of the chord of the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ whose mid point is $\left(1, \frac{2}{5}\right)$ is equal to:

[JEE Mains-2024]

- (A) $\frac{\sqrt{1691}}{5}$ (B) $\frac{\sqrt{2009}}{5}$ (C) $\frac{\sqrt{1741}}{5}$ (D) $\frac{\sqrt{1541}}{5}$

[Ans. : (A)]

DIBY-30

In triangle ABC , base BC is fixed. Then prove that the locus of vertex A such that $\tan B + \tan C = \text{Constant}$ is parabola.

DIBY-31

Prove that the line $px + qy + r = 0$ touches the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ if $a^2p^2 + b^2q^2 = r^2$

DIBY-32

Length of the latus-rectum of the ellipse represented by $x = 3(\cos t + \sin t)$, $y = 4(\cos t - \sin t)$; is given by

- (A) $\frac{5\sqrt{2}}{2}$ (B) $\frac{7\sqrt{2}}{2}$ (C) $\frac{9\sqrt{2}}{2}$ (D) $\sqrt{2}$

[Ans. : (C)]

DIBY-33

A common tangent to $9x^2 + 16y^2 = 144$, $y^2 = x - 4$ and $x^2 + y^2 - 12x + 32 = 0$ is

- (A) $y = 3$ (B) $y = -3$ (C) $x = 4$ (D) $x = -4$

[Ans. : (c)]

DIBY-34

The locus of the mid point of the line segment joining the point $(4, 3)$ and the points on the ellipse $x^2 + 2y^2 = 4$ is an ellipse with eccentricity: [JEE Mains-2022]

- (A) $\frac{\sqrt{3}}{2}$ (B) $\frac{1}{2\sqrt{2}}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{1}{2}$

[Ans. : (c)]

DIBY-35

Let a tangent to the curve $9x^2 + 16y^2 = 144$ intersect the coordinate axes at the points A and B . Then, the minimum length of the line segment AB is _____

[JEE Mains-2023]

[Ans. : 7]

DIBY-36

If the line $x - 2y = 12$ is tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2}$ at the point $\left(3, \frac{-9}{2}\right)$, then the length of the latus rectum of the ellipse is **[JEE Mains-2019]**

- (A) $12\sqrt{2}$ (B) 9 (C) $8\sqrt{3}$ (D) 5

[Ans.: (B)]

DIBY-37

The area (in sq. units) of the quadrilateral formed by the tangents at the end points of the latera recta to the ellipse $\frac{x^2}{9} + \frac{y^2}{5} = 1$ is **[JEE Mains-2016]**

- (A) $\frac{27}{2}$ (B) 27 (C) $\frac{27}{4}$ (D) 18

[Ans.: (B)]



DIBY-38

Let $A(\alpha, 0)$ and $B(0, \beta)$ be the points on the line $5x + 7y = 50$. Let the point P divide the line segment AB internally in the ratio $7 : 3$. Let $3x - 25 = 0$ be a directrix of the ellipse $E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the corresponding focus be S . If from S , the perpendicular on the x -axis passes through P , then the length of the latus rectum of E is equal to:

[JEE Mains-2024]

- (A) $\frac{25}{3}$ (B) $\frac{25}{9}$ (C) $\frac{32}{5}$ (D) $\frac{32}{9}$

[Ans. : (C)]

DIBY-39

A line passing through the point $P(\sqrt{5}, \sqrt{5})$ intersects the ellipse $\frac{x^2}{36} + \frac{y^2}{25} = 1$ at A and B such that $(PA) \cdot (PB)$ is maximum. Then $5(PA^2 + PB^2)$ is equal to:

[JEE Mains-2025]

- (A) 218 (B) 377 (C) 290 (D) 338

[Ans. : (D)]

DIBY-40

Let $P\left(\frac{2\sqrt{3}}{\sqrt{7}}, \frac{6}{\sqrt{7}}\right)$, Q , R and S be four points on the ellipse $9x^2 + 4y^2 = 36$. Let PQ and RS be mutually perpendicular and pass through the origin. If $\frac{1}{(PQ)^2} + \frac{1}{(RS)^2} = \frac{p}{q}$, where p and q are coprime, then $p + q$ is equal to **[JEE Mains-2023]**

- (A) 143 (B) 137 (C) 157 (D) 147

[Ans.: (C)]

DIBY-41

The centre of a circle C is at the centre of the ellipse $E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $a > b$. Let C pass through the foci F_1 and F_2 of E such that the circle C and the ellipse E intersect at four points. Let P be one of these four points. If the area of the triangle PF_1F_2 is 30 and the length of the major axis of E is 17, then the distance between the foci of E is: **[JEE Mains-2025]**

- (A) 26 (B) 13 (C) 12 (D) $\frac{13}{2}$

[Ans.: (B)]

DIBY-42

Let the ellipse $3x^2 + py^2 = 4$ pass through the centre C of the circle $x^2 + y^2 - 2x - 4y - 11 = 0$ of radius r . Let f_1, f_2 be the focal distances of the point C on the ellipse. Then $6f_1f_2 - r$ is equal to **[JEE Mains-2025]**

- (A) 74 (B) 68 (C) 70 (D) 78

[Ans. : (C)]

DIBY-43

Let the product of the focal distances of the point $(\sqrt{3}, \frac{1}{2})$ on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, (a > b)$, be $\frac{7}{4}$. Then the absolute difference of the eccentricities of two such ellipses is **[JEE Mains-2025]**

- (A) $\frac{3-2\sqrt{2}}{2\sqrt{3}}$ (B) $\frac{1-2\sqrt{2}}{\sqrt{3}}$ (C) $\frac{3-2\sqrt{2}}{3\sqrt{2}}$ (D) $\frac{1-\sqrt{3}}{\sqrt{2}}$

[Ans. : (A)]



*It is never about **end result**
It's always about the **Journey***

THANKYOU

#future **IITians**